

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech Theory Examination (Nov-Dec 2017)****MA202 Engineering Mathematics III****Date: 20.11.2017, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper contains two sections. Each section has three questions.
2. Section I and II must be attempted in separate answer sheets. Each section carries 35 marks.
3. Figures to the right indicate marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q - 1 Multiple Choice Questions****[07]**

- 1) The period of $\cos t$ is _____.
(i) 2π (ii) π (iii) $\frac{2\pi}{5}$ (iv) $\frac{\pi}{5}$
- 2) If $f(x) = x$; $-\pi < x < \pi$ and $f(x + 2\pi) = f(x)$, then $f(x)$ is _____.
(i) an even function (ii) an odd function (iii) neither even nor odd (iv) none of these
- 3) The Laplace transform of e^{-3t} is _____.
(i) $\frac{1}{s+3}$ (ii) $\frac{1}{s-3}$ (iii) $\frac{2}{s+3}$ (iv) $\frac{2}{s-3}$
- 4) The inverse Laplace transforms of $\frac{s}{s^2+a^2}$ is _____.
(i) $\sin at$ (ii) $\cos at$ (iii) $\sin hat$ (iv) $\cos at$
- 5) One of the roots of an equation $x^4 + 5x^3 + 4 = 0$ is _____.
(i) 0 (ii) 1 (iii) -1 (iv) 4
- 6) A biquadratic equation has _____ roots.
(i) 3 (ii) 4 (iii) 1 (iv) 2
- 7) An equation of the form $x^3 - 3x^2 + 2x - 3 = 0$ is called _____ equation.
(i) linear (ii) quadratic (iii) cubic (iv) biquadratic

Q - 2 Attempt any *four* out of six**[16]**

- 1) Find the Fourier series of the periodic function $f(x) = \frac{1}{2}(\pi - x)$; $0 \leq x \leq 2\pi$.
Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.
- 2) Express $f(x) = x^2$ as a Fourier series for all values of x from $-\pi$ to π .
- 3) Find the Laplace transforms of: (i) $e^{-t} \cos t$ (ii) $\sin t \cos t$.

- 4) Using the convolution theorem, find $L^{-1}\left\{\frac{1}{s^2(s-1)}\right\}$.
- 5) Solve the differential equation $y'' + 5y' + 6y = 0$ and $y(0) = 0$, $y'(0) = 1$ using Laplace transform.
- 6) Find the solution of the equation $x^3 - 18x - 35 = 0$ by Cardan's method.

Q – 3 (a) Attempt any two out of three

[06]

- 1) Find the half range sine series of the function $f(x) = c$; $0 < x < \pi$.
- 2) Find the Laplace transform of the periodic triangular wave function of period $2a$ defined as $f(t) = \begin{cases} k, & 0 \leq t < a \\ -k, & a \leq t < 2a \end{cases}$.
- 3) Find the inverse Laplace transform of the function $F(s) = \frac{1}{s^2+s}$.

Q – 3(b) Attempt any two out of three

[06]

- 1) Find the Fourier series of the periodic function $f(x) = x$; $-1 < x < 1$.
- 2) Find the Laplace transform of $f(t) = \frac{1-\cos 2t}{t}$.
- 3) Solve the equation $x^4 + 2x^3 - 7x^2 - 8x + 12 = 0$ by Ferrari Method.

SECTION – II

Q – 4 Multiple Choice Questions

[07]

- 1) If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then $\text{div } \vec{r}$ equal to
(a) 3 (b) r (c) 0 (d) -r
- 2) The vector defined by $\vec{F} = e^x \sin y\hat{i} + e^x \cos y\hat{j}$ is
(a) rotational (b) irrotational (c) solenoidal (d) conservative
- 3) Stokes theorem connects
(a) a line integral and surface integral
(b) a surface integral and a volume integral
(c) a line integral and a volume integral
(d) gradient of a function and its surface integral
- 4) The vector field $\vec{F} = x\hat{i} - y\hat{j}$
(a) divergence free, but not irrotational
(b) irrotational, but not divergence free
(c) divergence free and irrotational
(d) neither divergence nor irrotational
- 5) A necessary and sufficient condition condition that line integral $\oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed curve C is that
(a) $\text{div } \vec{f} = 0$ (b) $\text{curl } \vec{f} = 0$ (c) $\text{div } \vec{f} \neq 0$ (d) $\text{curl } \vec{f} \neq 0$
- 6) Let $f(x,y,z)=c$ represent the equation of the surface. Then unit normal vector to this surface is
(a) $\frac{\text{grad } f}{|\text{grad } f|}$ (b) $\text{grad } f$ (c) $\text{div}(\text{grad } f)$ (d) $\text{curl}(\text{grad } f)$

- 7) The expression $\text{curl}(\text{grad } f)$, where f is a scalar function, is
 (a) equal to $\nabla^2 f$ (b) equal to $\text{div}(\text{grad } f)$ (c) a scalar of zero magnitude
 (d) a vector of zero magnitude

Q -5 Attempt any four out of six

[16]

- 1) Show that the frequency of free vibrations in a closed electrical circuit with inductance L and capacity C in series is $\frac{30}{\pi\sqrt{LC}}$ cycles/minute.
- 2) Prove that $\vec{F} = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + 3xz^2\hat{k}$ is irrotational and find its scalar potential.
- 3) Evaluate $\iint \vec{F} \cdot \hat{n} \, ds$ where, $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$ taken over the rectangular parallelepiped $x=0, y=0, z=0, x=a, y=b, z=c$.
- 4) Verify Stokes theorem for $\vec{F} = xy^2\hat{i} + y\hat{j} + z^2x\hat{k}$ for the surface of a rectangular lamina bounded by $x=0, y=0, x=1, y=2, z=0, z=1$.
- 5) Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at the point $(2, -1, 2)$ along the direction normal to the surface $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$.
- 6) Differential equation for the elastic curve of the beam is $\frac{d^2y}{dx^2} + \frac{P}{EI}y = -\frac{W}{2EI}x$, given that $y(0) = 0$; at $x = l/2, \frac{dy}{dx} = 0$ then Prove that

$$y = \frac{W}{2P} \left(\frac{\sin nx}{n \cos \frac{nl}{2}} - x \right)$$

Q -6 Attempt any two out of three

[06]

(a)

- 1) Find the angle between the tangents to the curves $x=t^2, y=2t, z=-t^3$ at the points $t=1$ and $t=-1$.
- 2) Prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$
- 3) Evaluate the line integral for the vector field $\vec{F} = x^2\hat{i} - xy\hat{j}$ from the origin O to the point $P(1,1)$ along the straight line OP .

Q -6 Attempt any one out of two

[06]

(b)

- 1) A body executes damped forced vibrations given by the equations $\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 64x = 128\cos 8t$. Solve the equation, when $x = \frac{1}{3}, \frac{dx}{dt} = 0$ at $t = 0$.
- 2) Verify Divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ over the cube bounded by the planes $x=0, y=0, x=2, y=2, z=0, z=2$

